Predictive Code-Switching Detection: A Universality-Focused Multitask Learning Approach

Jae Hun Cho
Northeastern University
Seattle, USA
cho.jae@northeastern.edu

Fengying Zeng
Northeastern University
Seattle, USA
zeng.fen@northeastern.edu

Xiaoyan Cai
Northeastern University
Seattle, USA
cai.xiaoyan@northeastern.edu

Abstract

We propose a predictive code-switching detection system that anticipates language switches *before* they occur in bilingual text. Unlike traditional detection approaches that identify switches after the fact, our model predicts at token position t whether token $t + 1$ will be a language switch. We employ multitask learning with two objectives: (1) binary switch prediction, and (2) switch duration classification (small: 1-2 tokens, medium: 3-6 tokens, large: 7+ tokens). Using the SwitchLingua dataset (420K samples, 12+ language pairs), we compare XLM-RoBERTa against mBERT, evaluating model *universality*—consistency of F1 scores across language pairs measured by low variance (σ).

1 Problem Description

Code-switching, the alternation between languages within a conversation, is a natural phenomenon among bilingual speakers. Traditional NLP approaches detect switches *after* they occur in complete sentences. However, real-time applications such as bilingual keyboards, chatbots, and translation tools require *predictive* capabilities—anticipating switches using only preceding context. This project addresses this gap by building a causal model that predicts upcoming language switches and their duration, with a focus on achieving consistent performance across diverse language pairs rather than optimizing for any single pair.

2 Background

In the work done by Khanuja et al. [1], they introduced GLUECoS, a benchmark for evaluating code-switched NLP tasks across English-Hindi and English-Spanish. They compared mBERT against cross-lingual word embeddings and fine-tuned mBERT on synthetically generated code-switched data. Although mBERT significantly outperformed baselines on syntactic tasks, performance on semantic tasks such as sentiment analysis remained considerably low.

In the work done by Aguilar et al. [2], they developed LinCE, a centralized benchmark covering four code-switched language pairs with standardized splits. They evaluated LSTM, ELMo, and mBERT baselines while introducing the Code-Mixing Index (CMI) to quantify switching intensity. However, their findings revealed that existing approaches are hardly generalizable, with performance varying by over 20 F1 points across language pairs.

In the work done by Zhang et al. [3], they addressed scalable token-level language identification for code-mixed text. They proposed a feed-forward neural network with a Viterbi decoder enforcing language transition constraints. Although the model achieved 800x speedup and 19.5% F1 improvement, performance degraded significantly on low-resource language pairs.

In the work done by Winata et al. [4], they conducted a systematic survey analyzing 400+ code-switching papers from two decades of NLP research. They categorized methods from rule-based to neural approaches and identified research gaps across tasks and languages. Notably, they found that predictive code-switching remains under-explored, with over 60% of datasets remaining private.

In the work done by Mager et al. [5], they addressed intra-word code-switching where morphemes from multiple languages appear within single tokens. They developed Segmental RNNs that jointly segment words and assign language tags at the subword level. Although the approach strongly outperformed baselines on morphologically mixed words, it requires expensive subword-level annotations limiting scalability.

3 Project Plan and Milestones

Week	Task	Deliverable
3-4	Data preprocessing	Parsed SwitchLingua dataset
5-6	Label engineering	Shifted labels for both tasks
7-8	Model architecture	Dual-head classifier + causal mask
9-10	Training pipeline	XLM-R & mBERT training
11-12	Hyperparameter tuning	Optimized model weights
13	Universality analysis	F1 variance across 12 pairs
14	Final paper	Complete research paper

References

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